Purpose

This is a step-by-step procedure to build/configure the online Intermediate Certificate Authority and handle all certificate tasks for the [INTERAL-ORG] environment, utilizing the trust established by the offline Root CA

In layman terms, the online Intermediate CA will represent the offline Root CA and conduct any certificate-related tasks for our environment.

Scope

In-Scope

- Installation of AD CS role as an Enterprise Subordinate
- Configuration of standardized Certification encryption for use by Windows and Non-Windows devices
- Exporting Certification Signing Request from Intermediate CA server

Out-of-Scope

- Joining the server onto [INTERNAL-ORG] domain

Requirements

Permissions: Domain/Local Adminstrator

Dedicated hardware/VM for Windows Server that will be online for certification

Software: Windows Server 2019 or higher

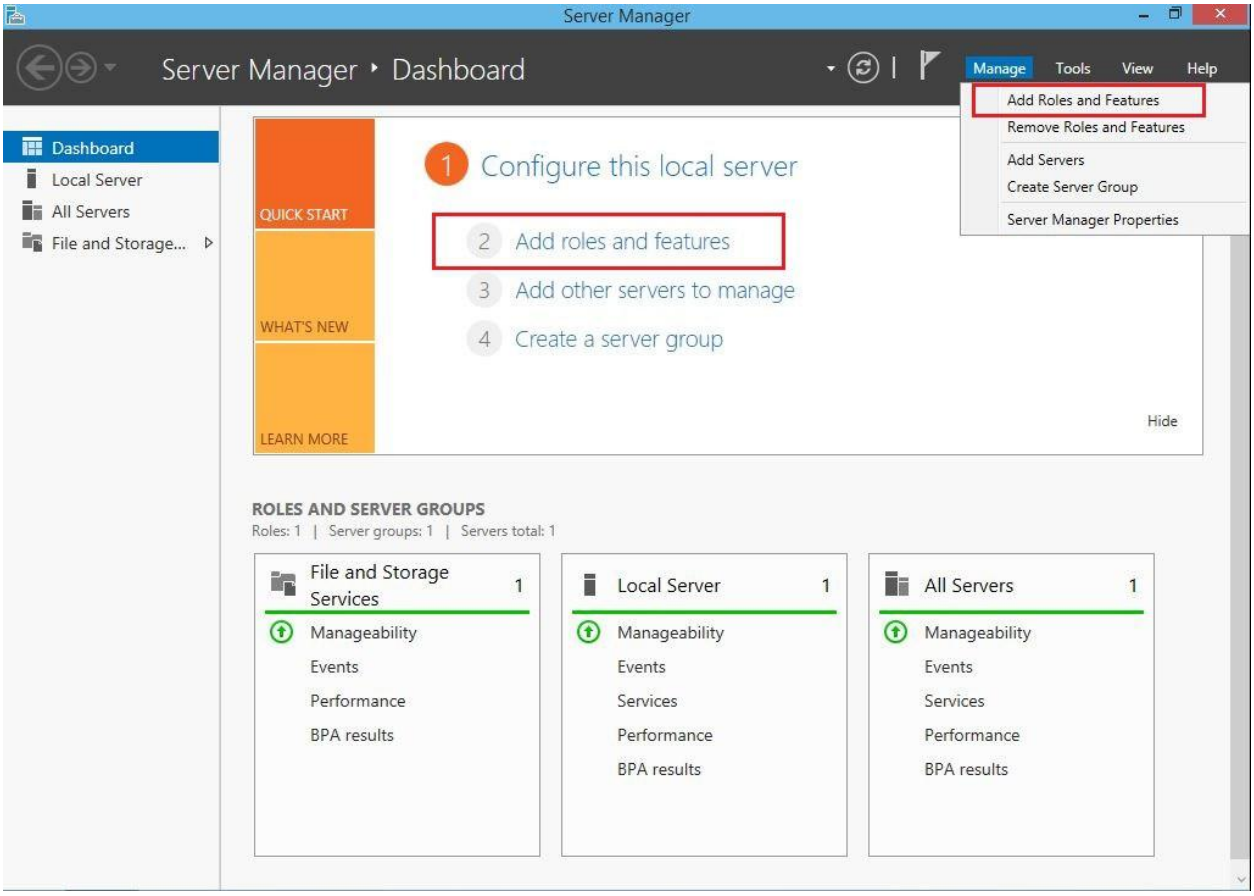
System must a FQDN (preferably INTERCA.[INTERNAL-ORG].local)

System must be joined to the domain

Procedure

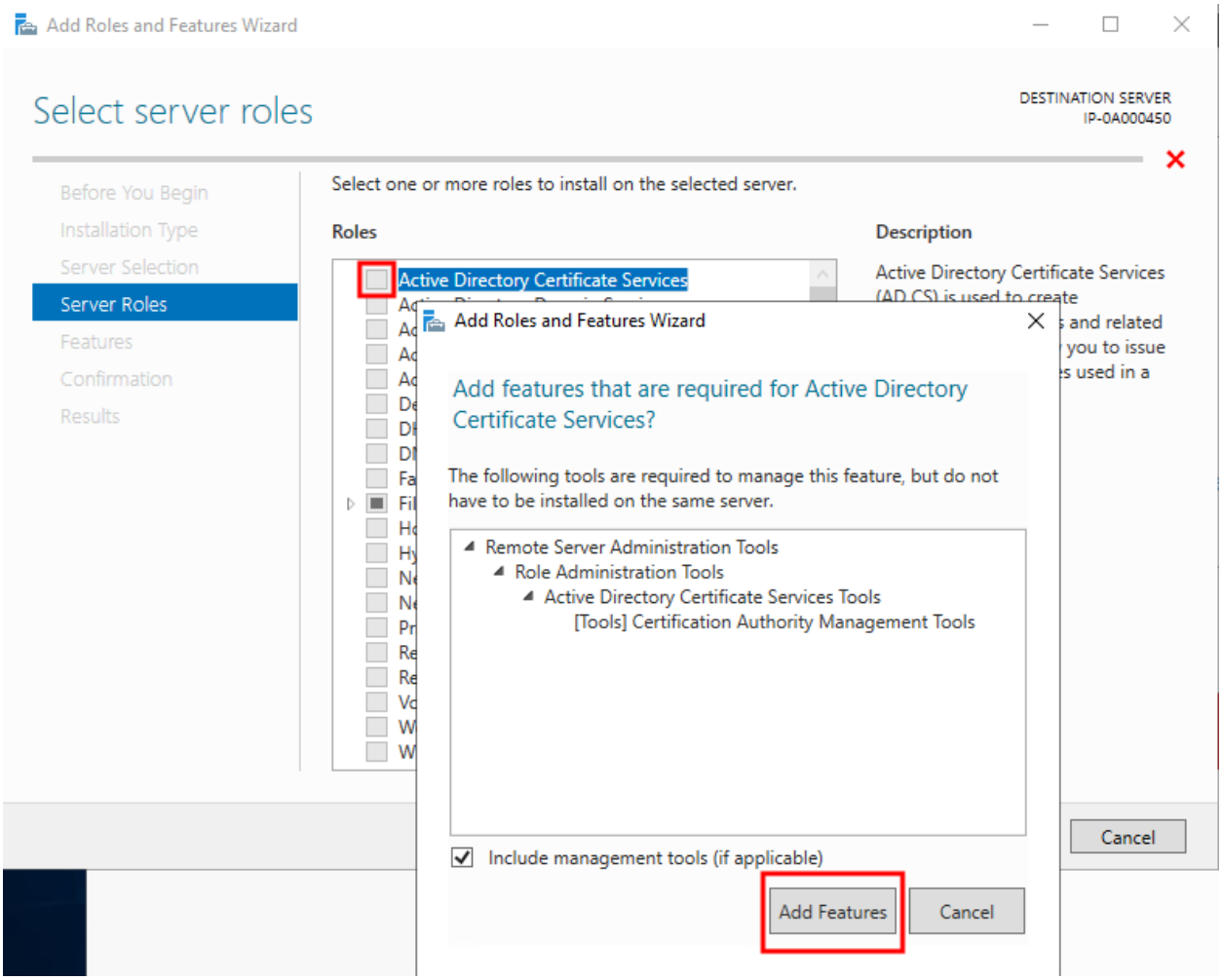
Intermediate CA Role Installation

1. Open **Server Manager -> Manage -> Add Roles and Features**



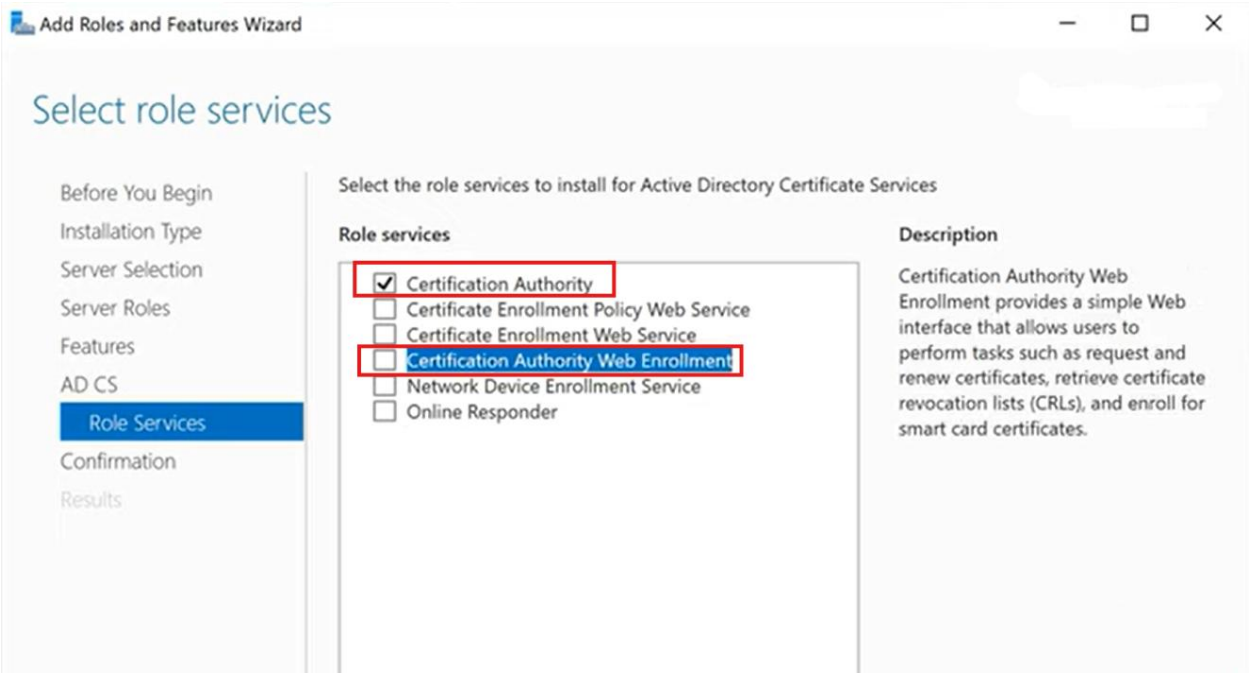
2. In the **Add Roles and Features Wizard**, accept defaults until **Server Roles**

3. **In Server Roles,** add the role **Active Directory Certificate Services** and click **Add Features**



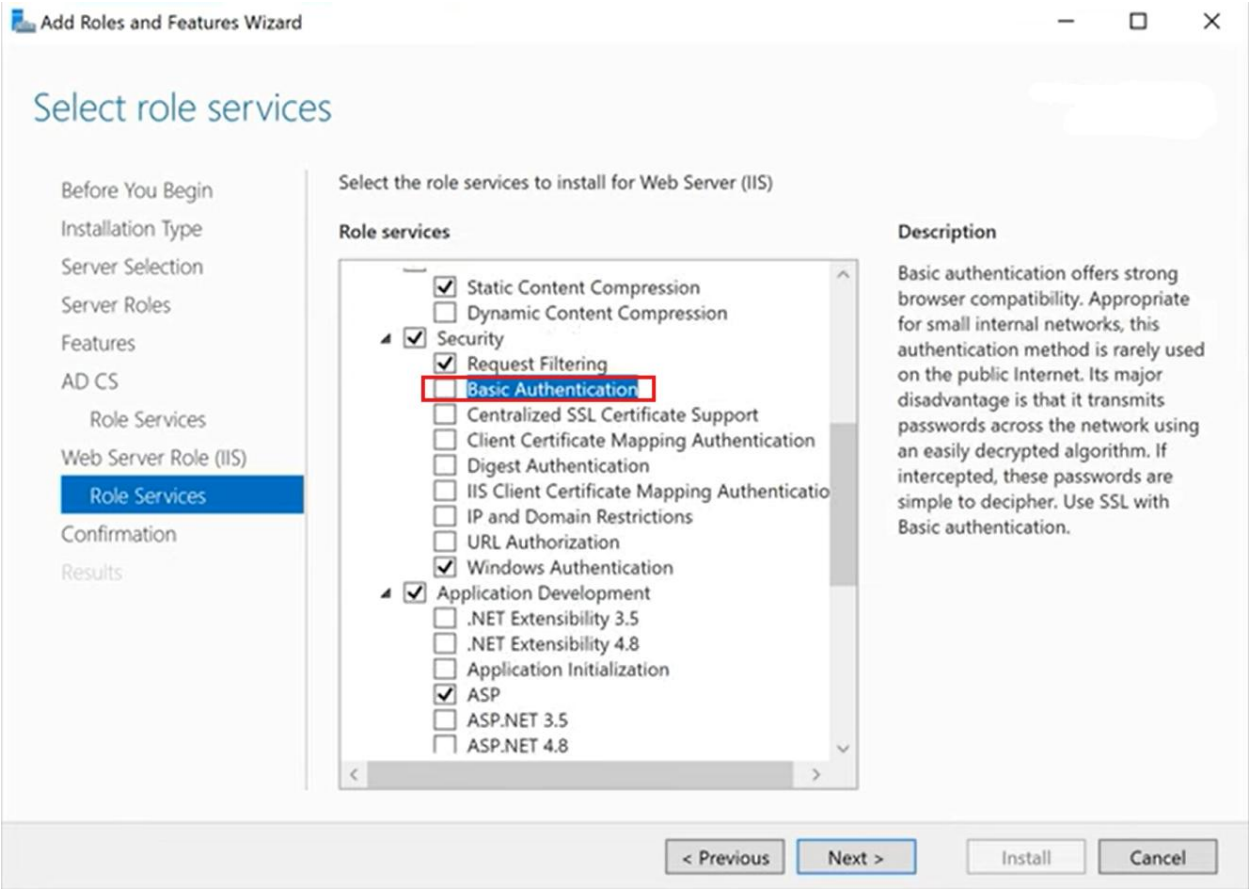
4. Accept all defaults and until **AD CS -> Role Services**

5. In **Role Services**, in addition to **Certificate Authority**, **check Certification Authority Web Enrollment** and **click Add Features**



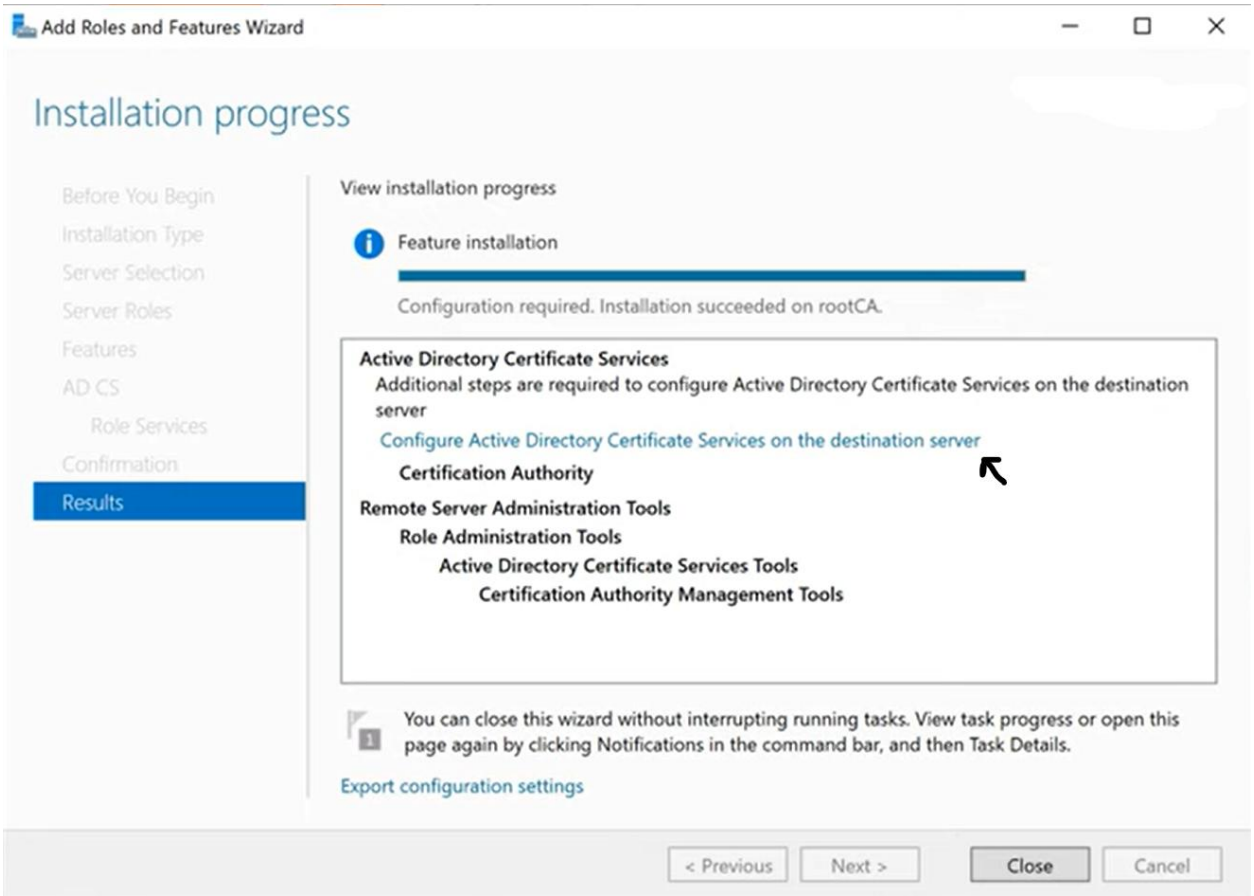
6. Accept all defaults and until **Web Server Roles (IIS) -> Role Services**

7. In **Role Services**, check **Basic Authentication** under **Web Sever -> Security**



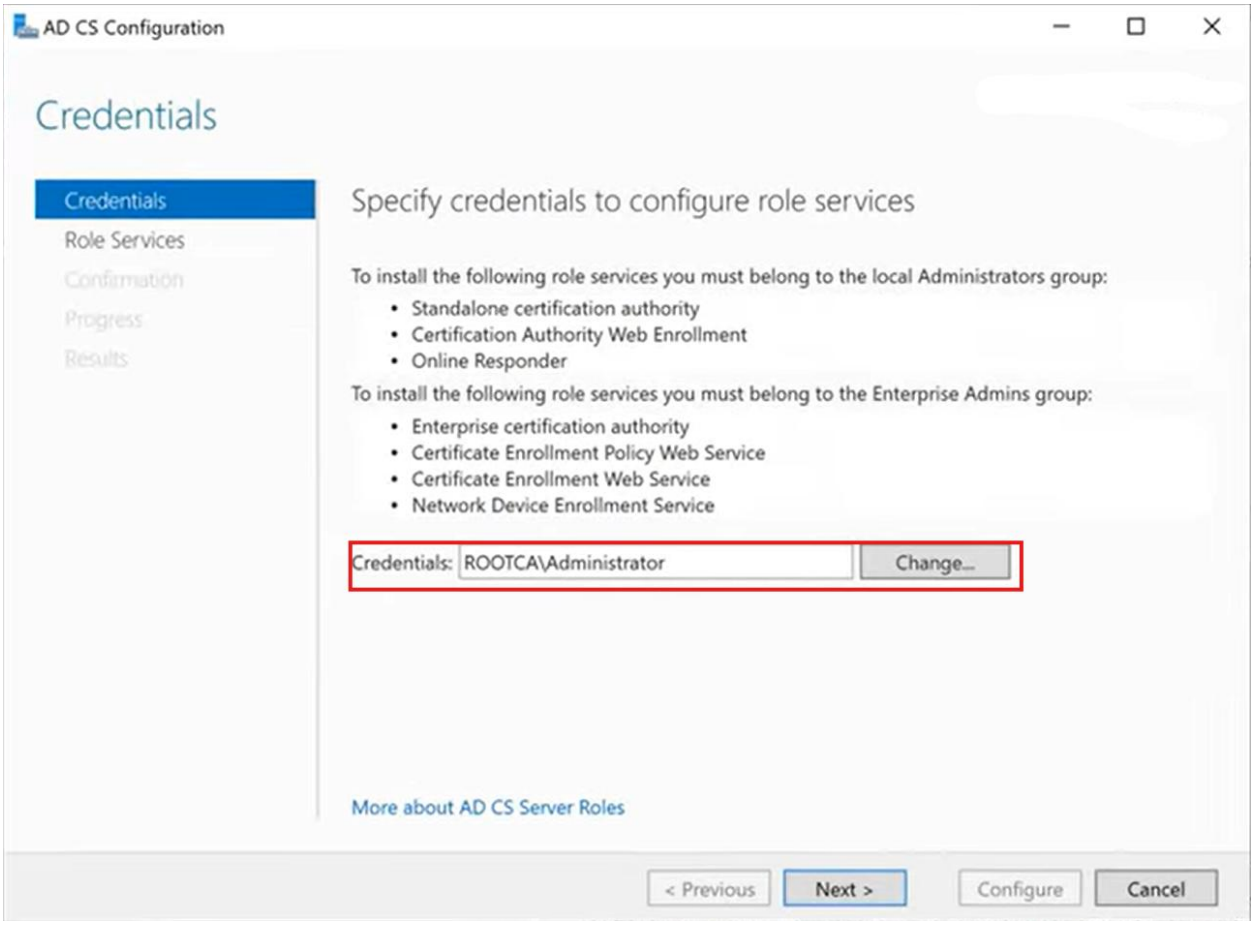
8. Click Next, Confirm configurations, and Install.

9. Once the installation is finished, on progress screen, click **Configure Active Directory Certificate Service on the destination server**

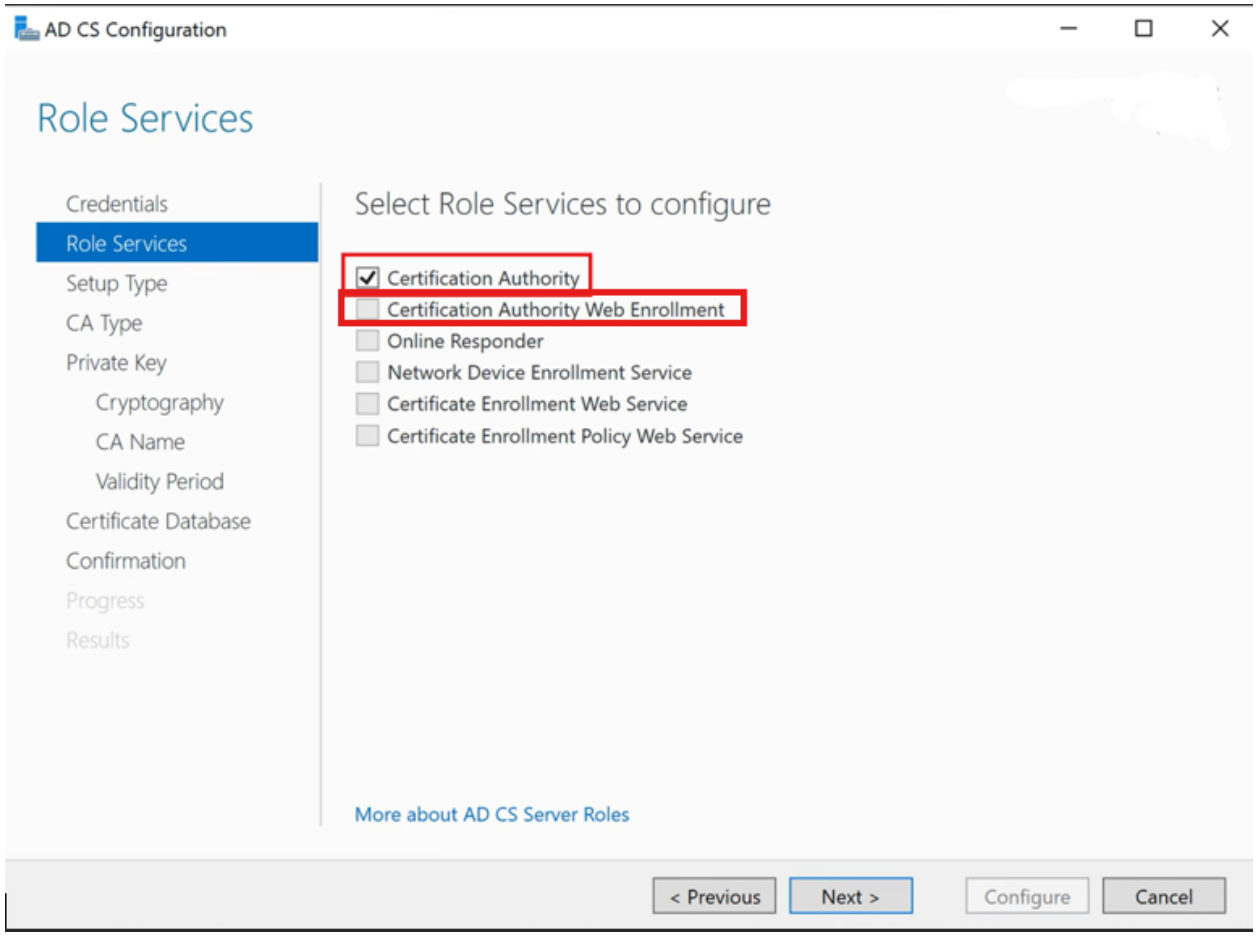


Post-Installation Role Configuration

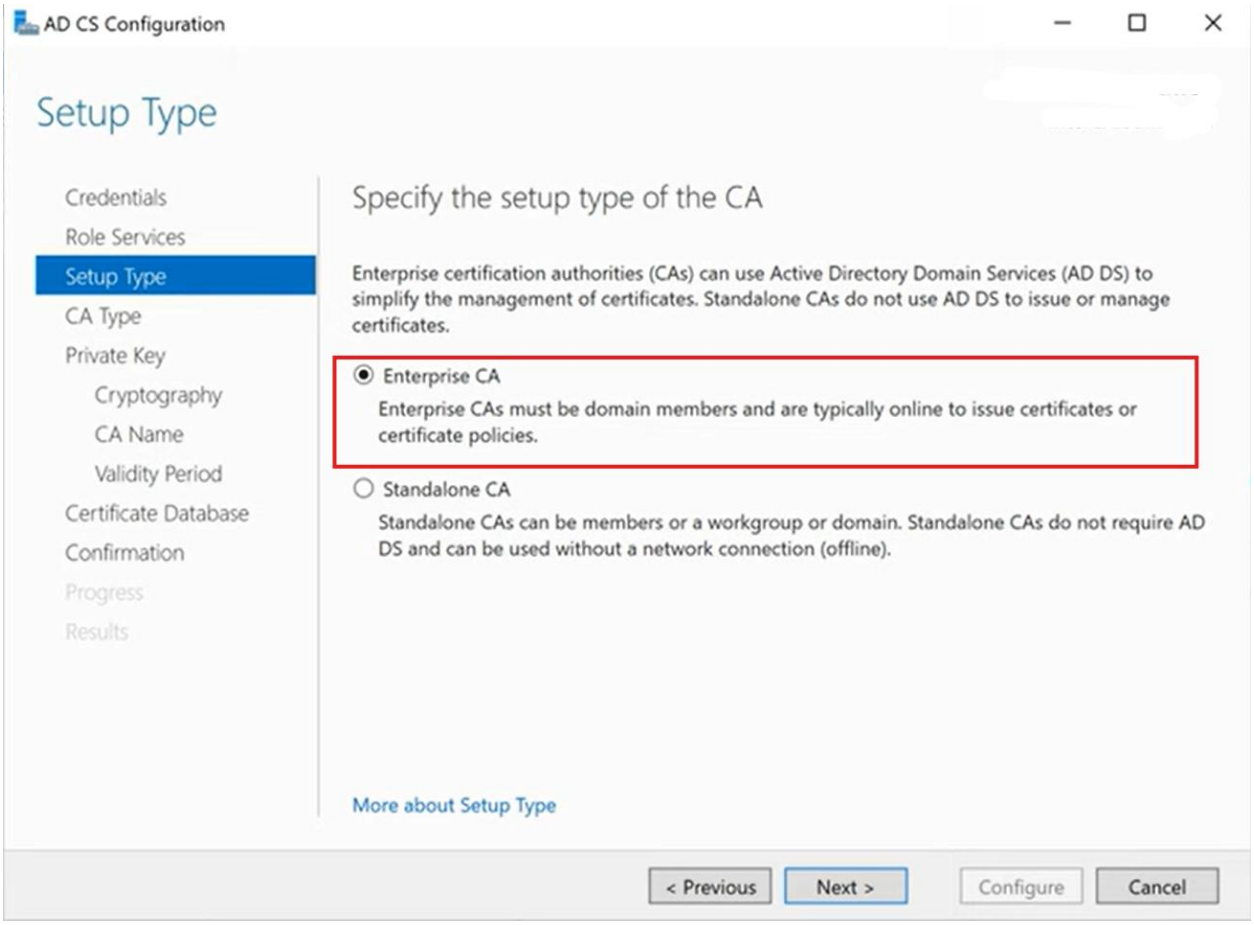
1. Ensure the **Credentials** are for the **Domain Administrator account** (following is an example)



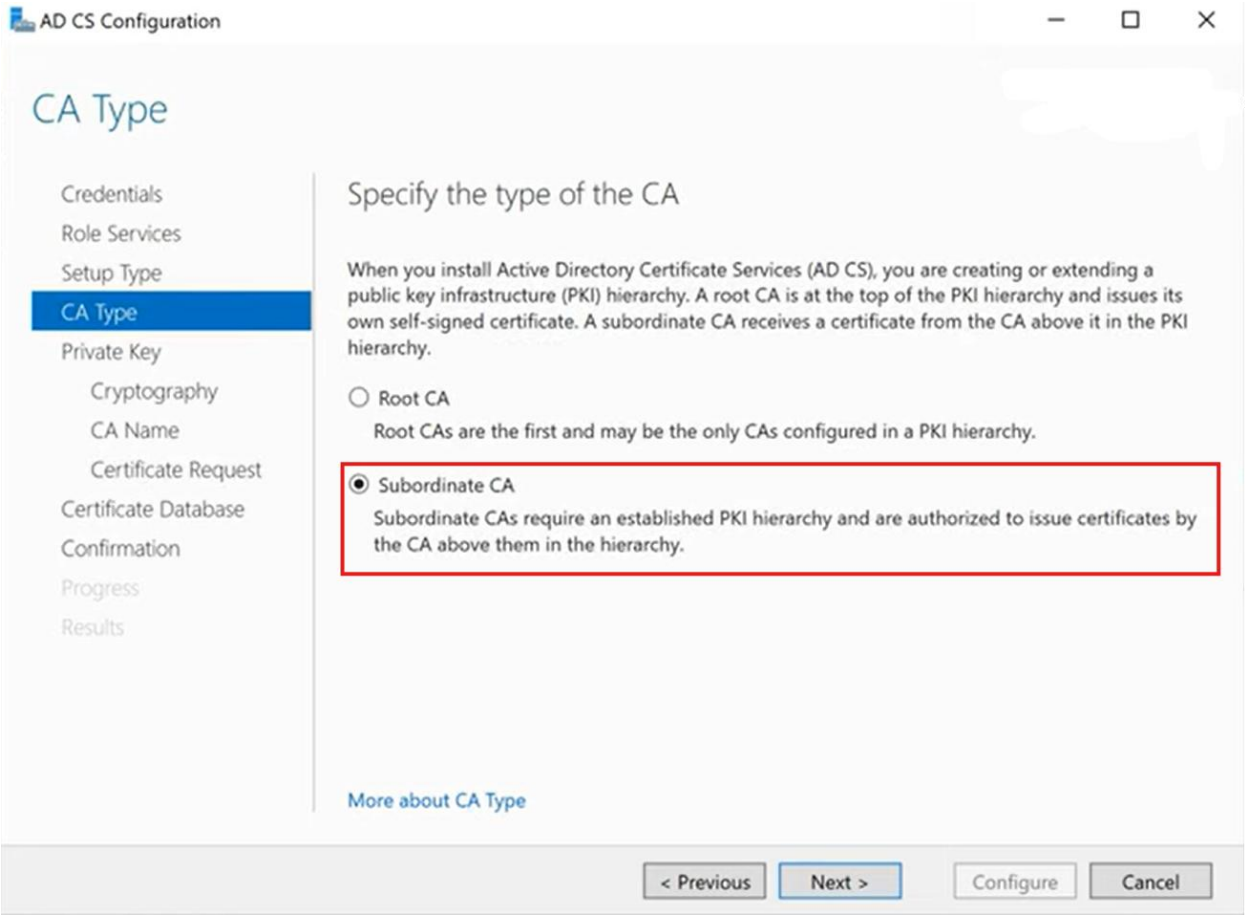
2. For **Role Services**, check **Certification Authority** and **Certification Authority Web Enrollment**



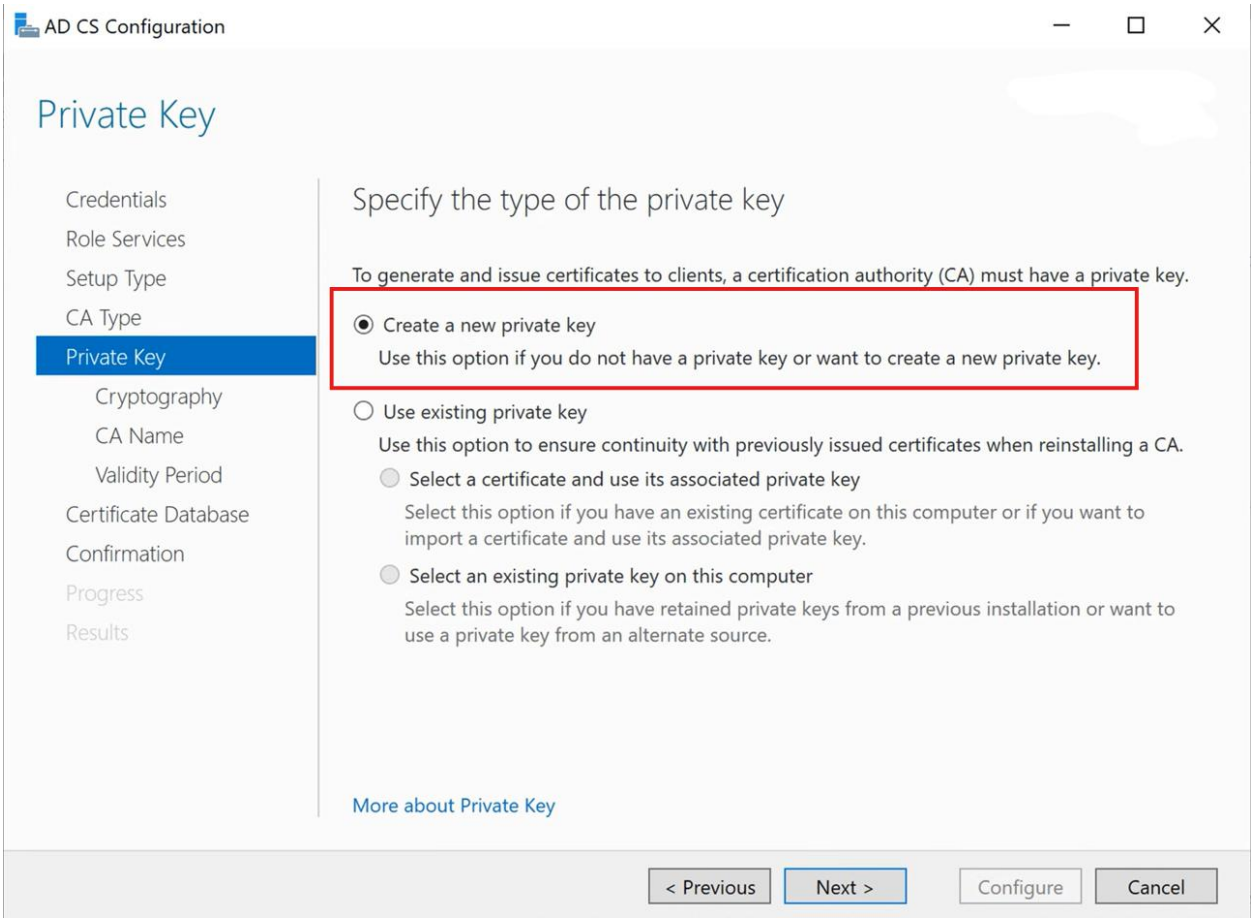
3. For **Setup Type**, select **Enterprise CA**



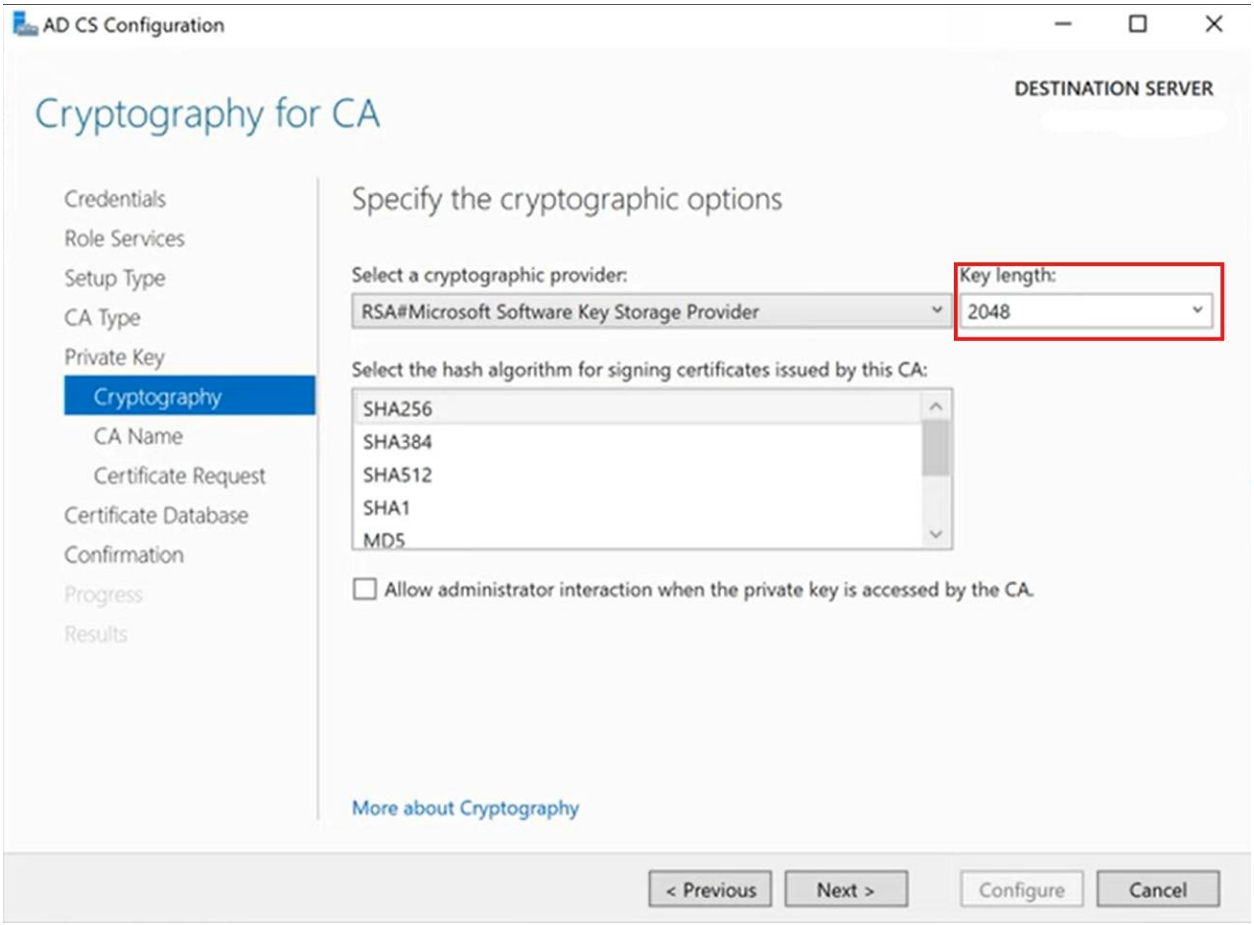
4. For **CA Type**, select **Subordinate CA**



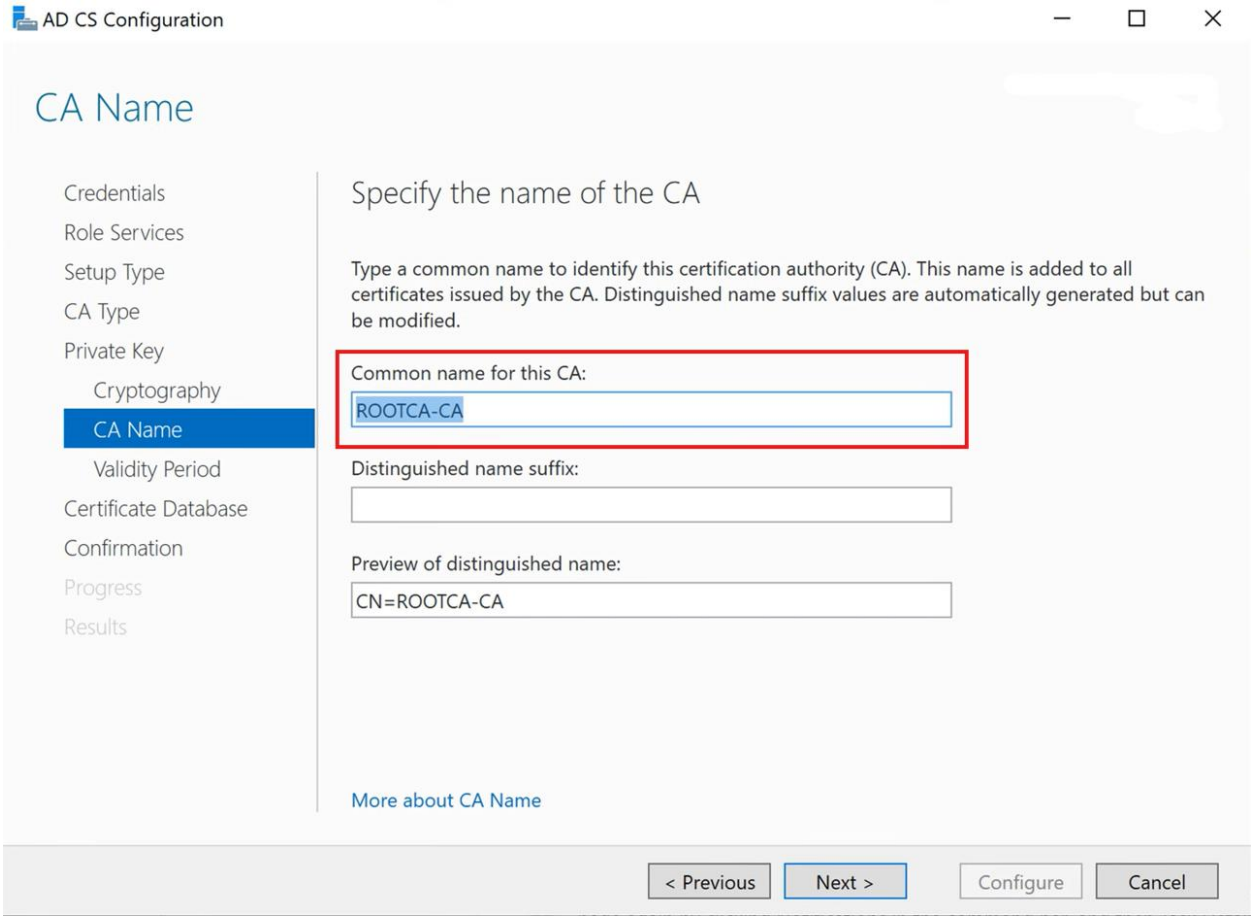
5. For **Private Key**, select **Create a new private key**



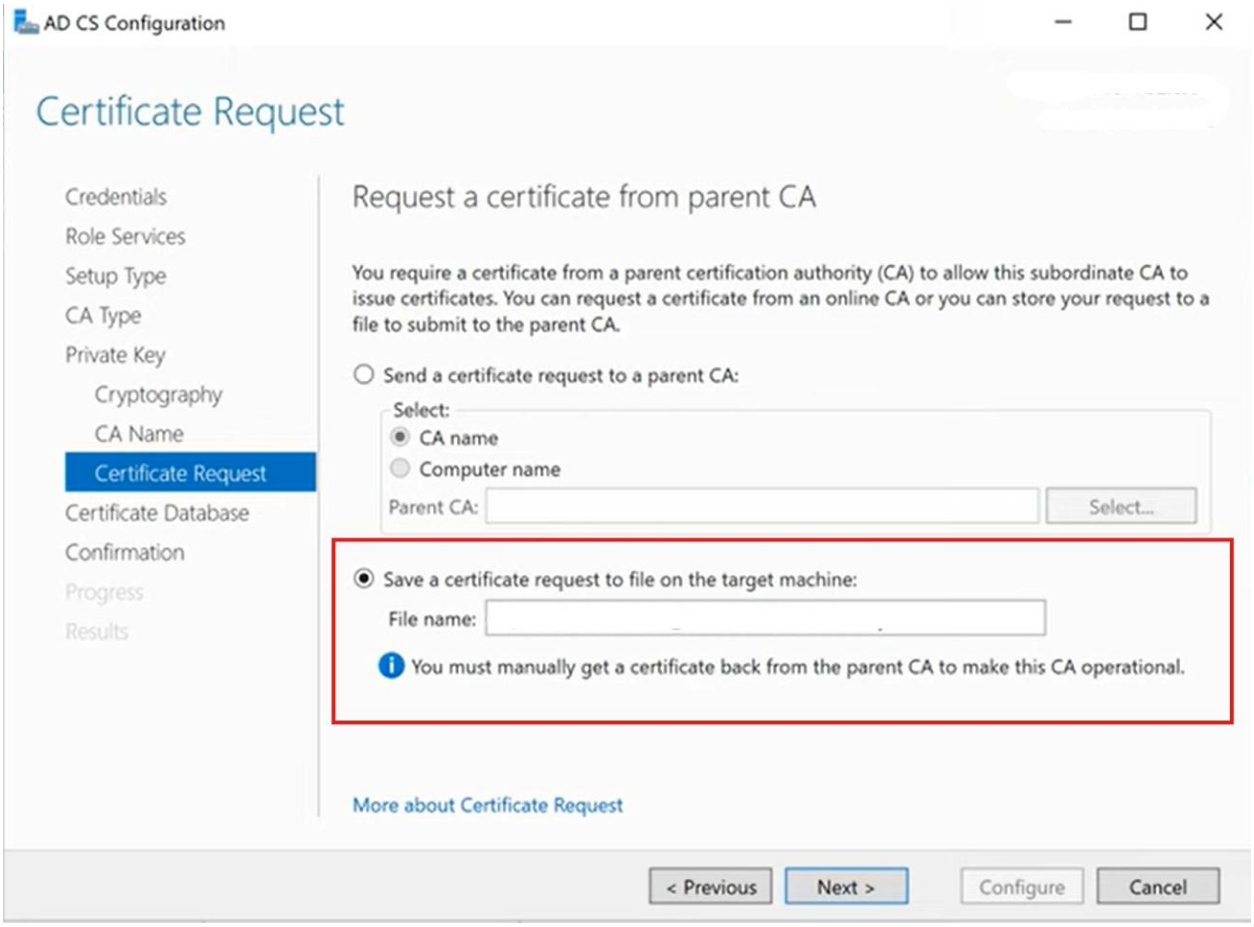
6. For **Cryptography**, select the key length **(needs to be compatible with all [INTERNAL-ORG] devices: Windows and Non-Windows /// Most likely 2048-bit)**



7. For **CA Name**, set the Common name as **[INTERNAL-ORG]-INTERCA-CA** (can be changed for better distinction but must match RootCA signing)



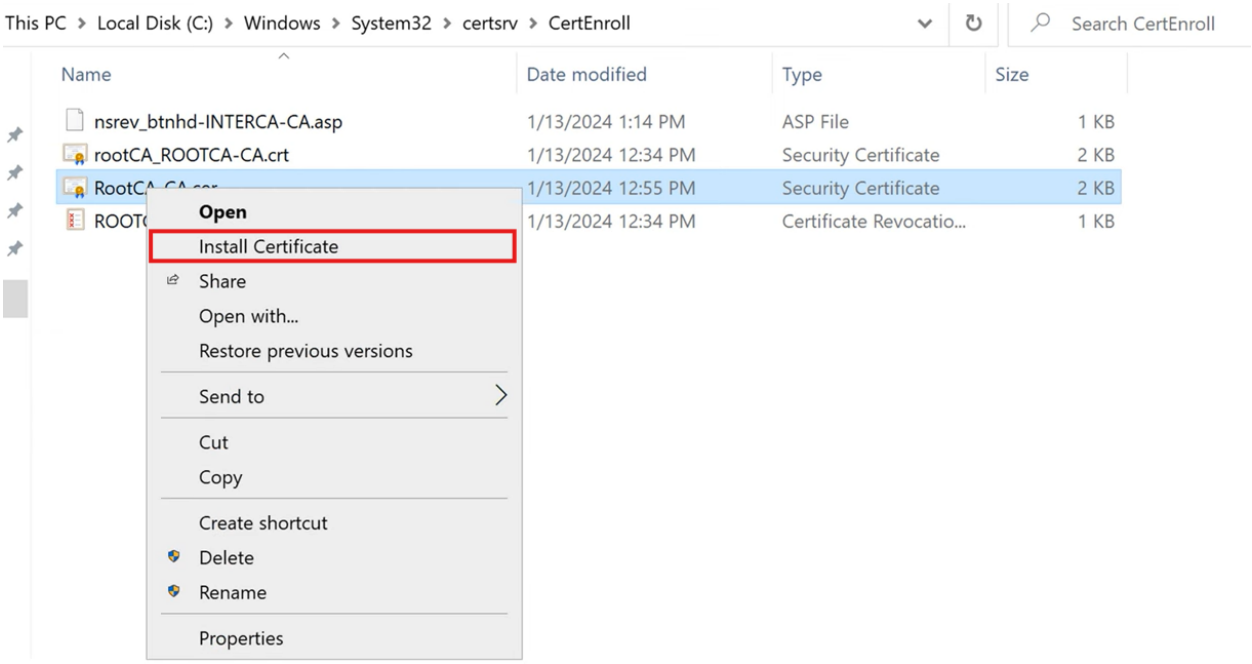
8. For **Certificate Request**, select **Save a certificate request to file on the target machine** and leave **File name** as default. (This option requires us to manually transfer the cert request from the Intermediate CA server to the Root CA server)



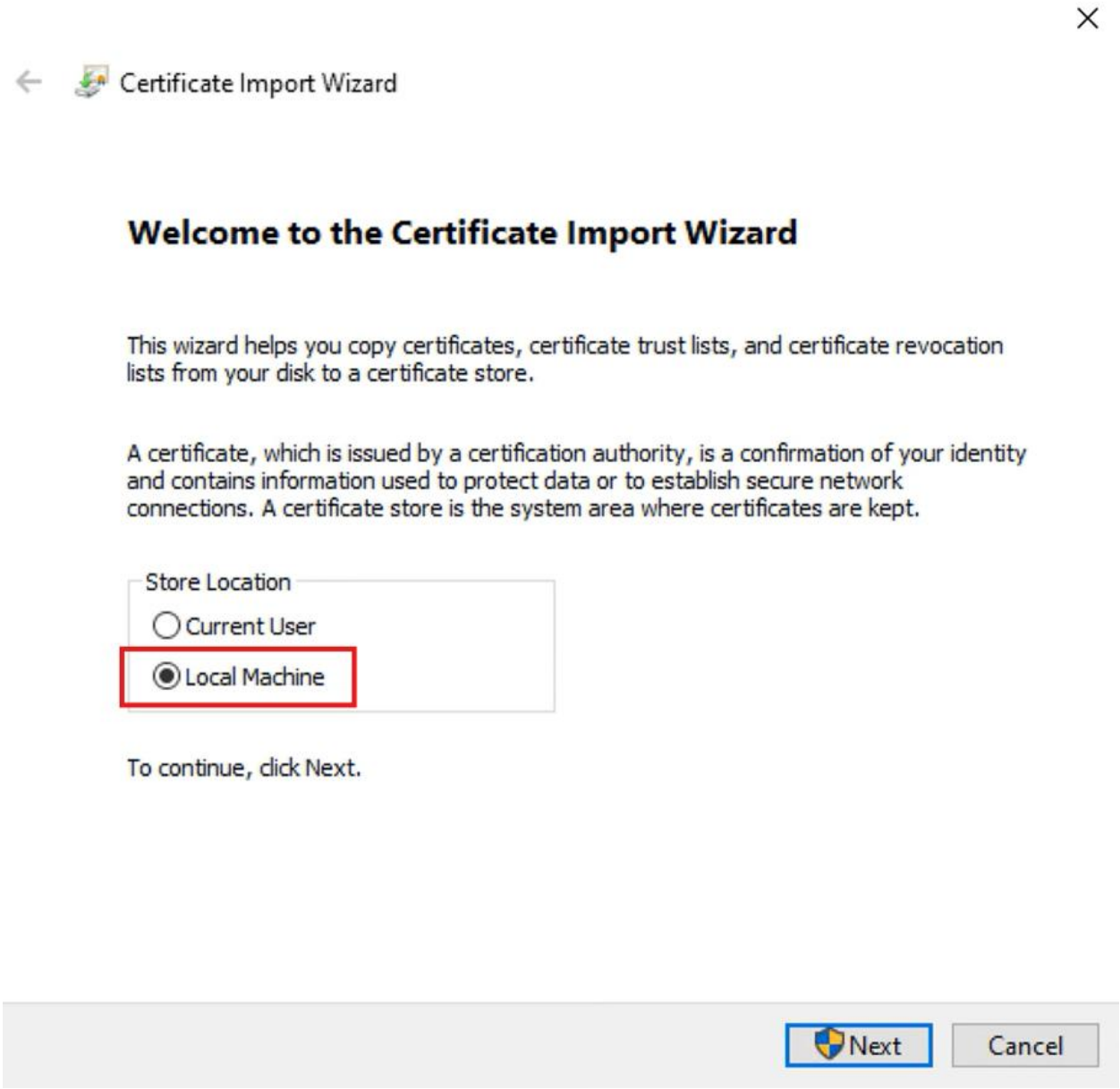
9. For **Certificate Database**, leave as default
10. Confirm configuration and hit **Configure**

Root CA Cert Installation

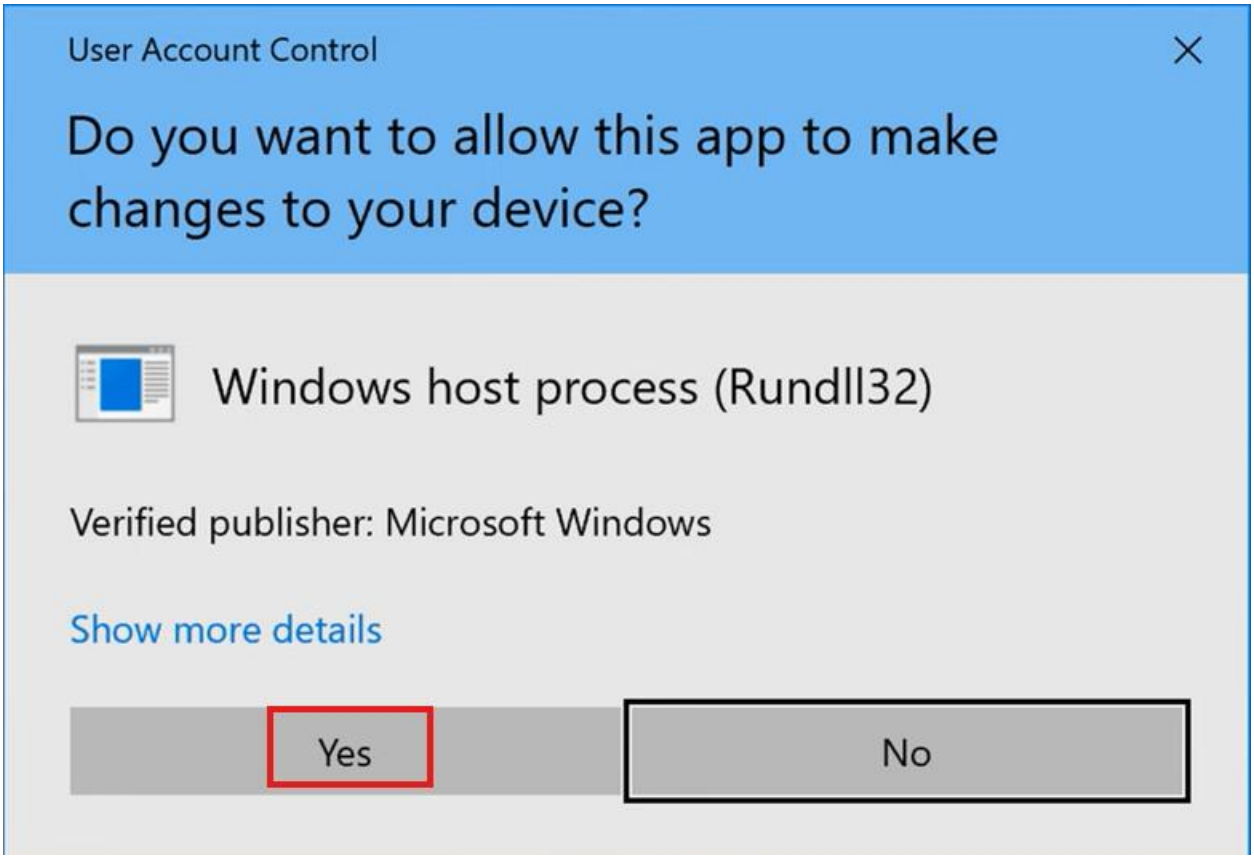
1. Mount the secure media containing the root certificate file (**.cer**) and the root revocation list file (**.crl**)
2. Transfer the root files to the Intermediate CA Server's file path **C:\Windows\System32\CertSrv\CertEnroll**
3. **Right-click** the **.cer** file and select **Install Certificate**



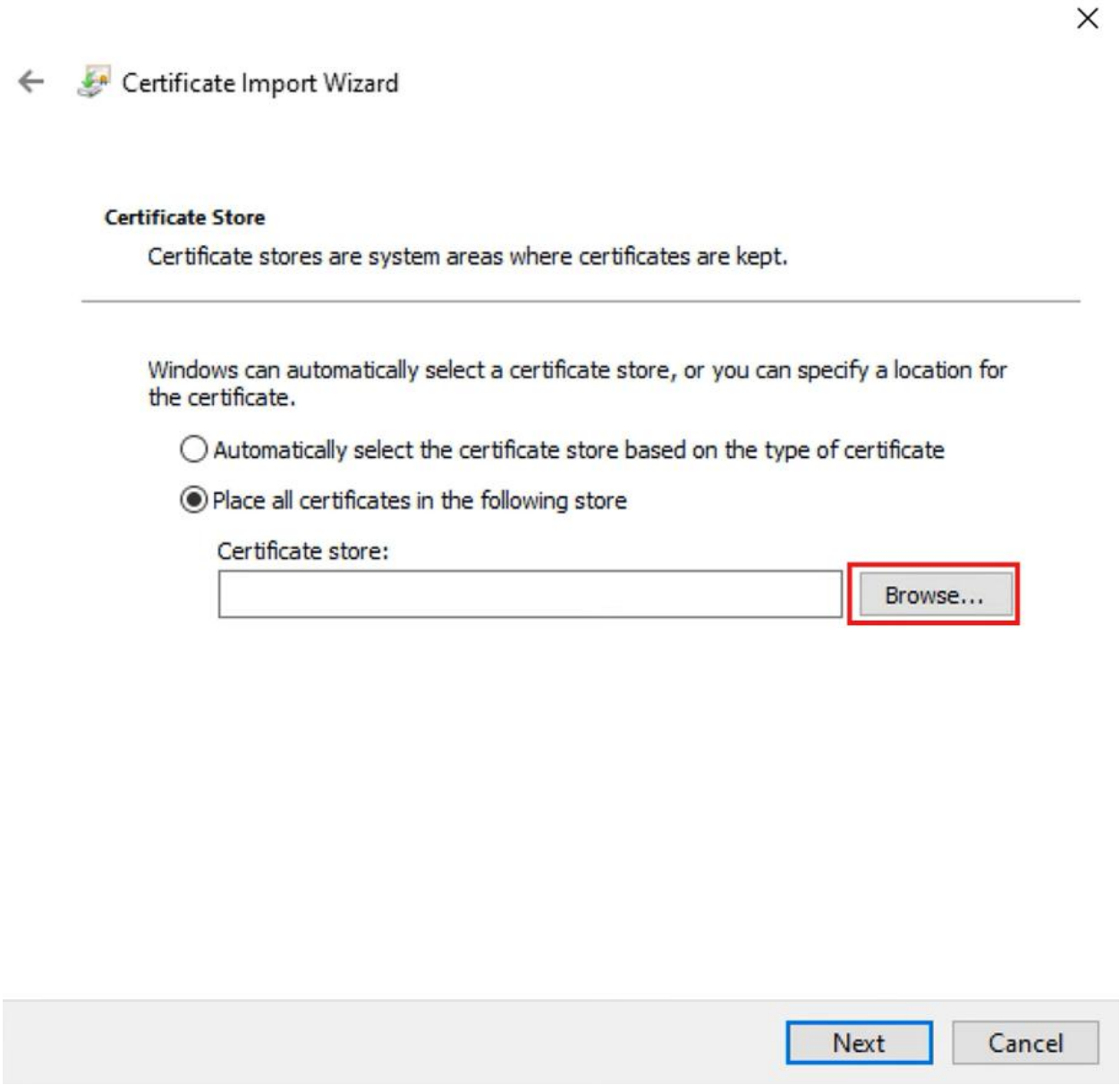
4. In **Certificate Import Wizard**, set the Store Location as **Local Machine**



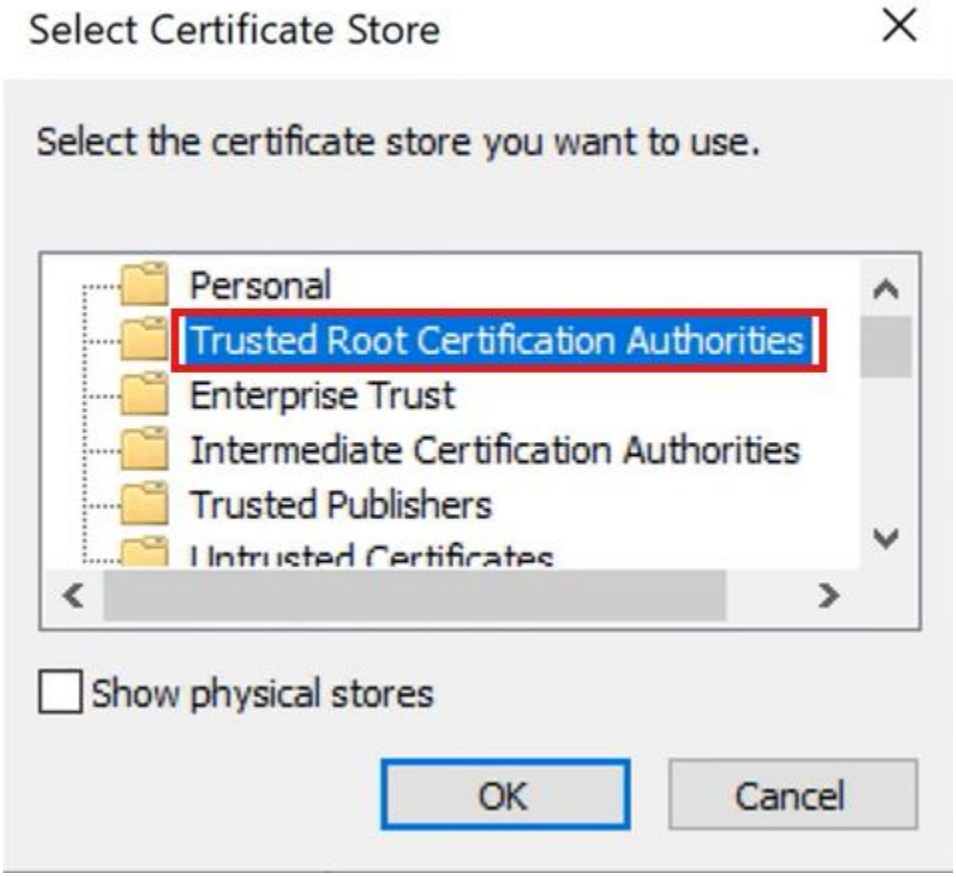
5. In the UAC popup, allow **Rundll32** to make changes



6. Select **Place all certificates in the following store** and click **Browse...**



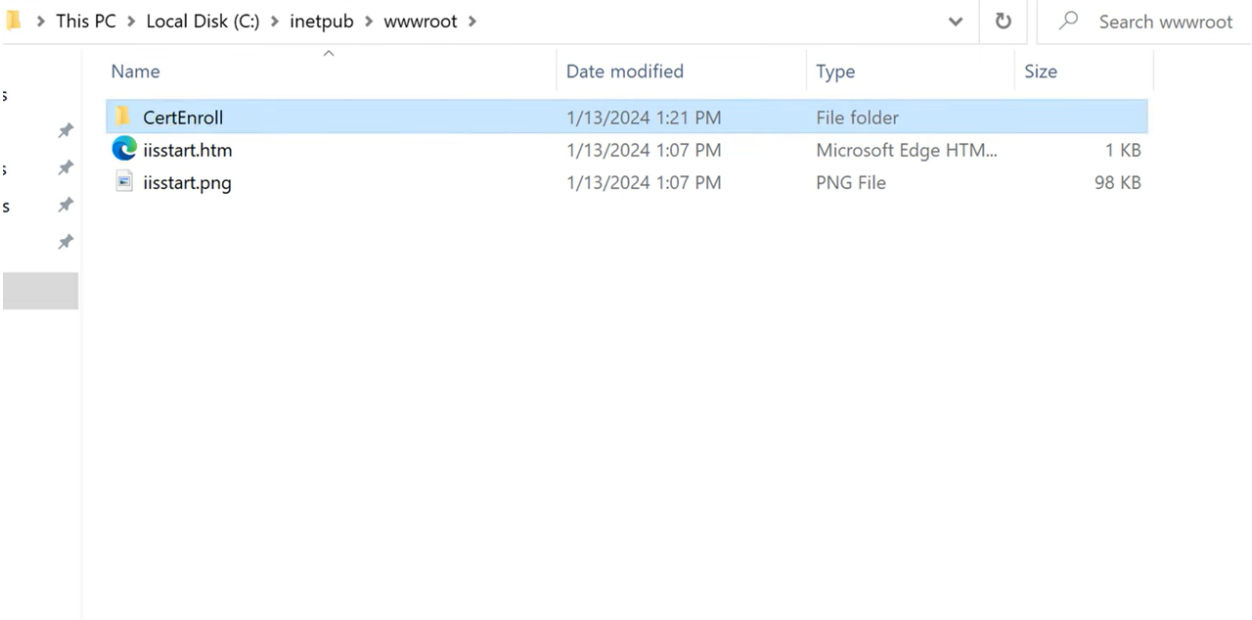
7. Set the Certificate Store as **Trusted Root Certification Authorities**



8. Click **Ok**, Click **Next**, Click **Finish.**
9. Repeat steps 3-8 for the **.crl** file (store in intermediate cert auth)

CertEnroll Folder Creation For Environment Certifications & IIS Config

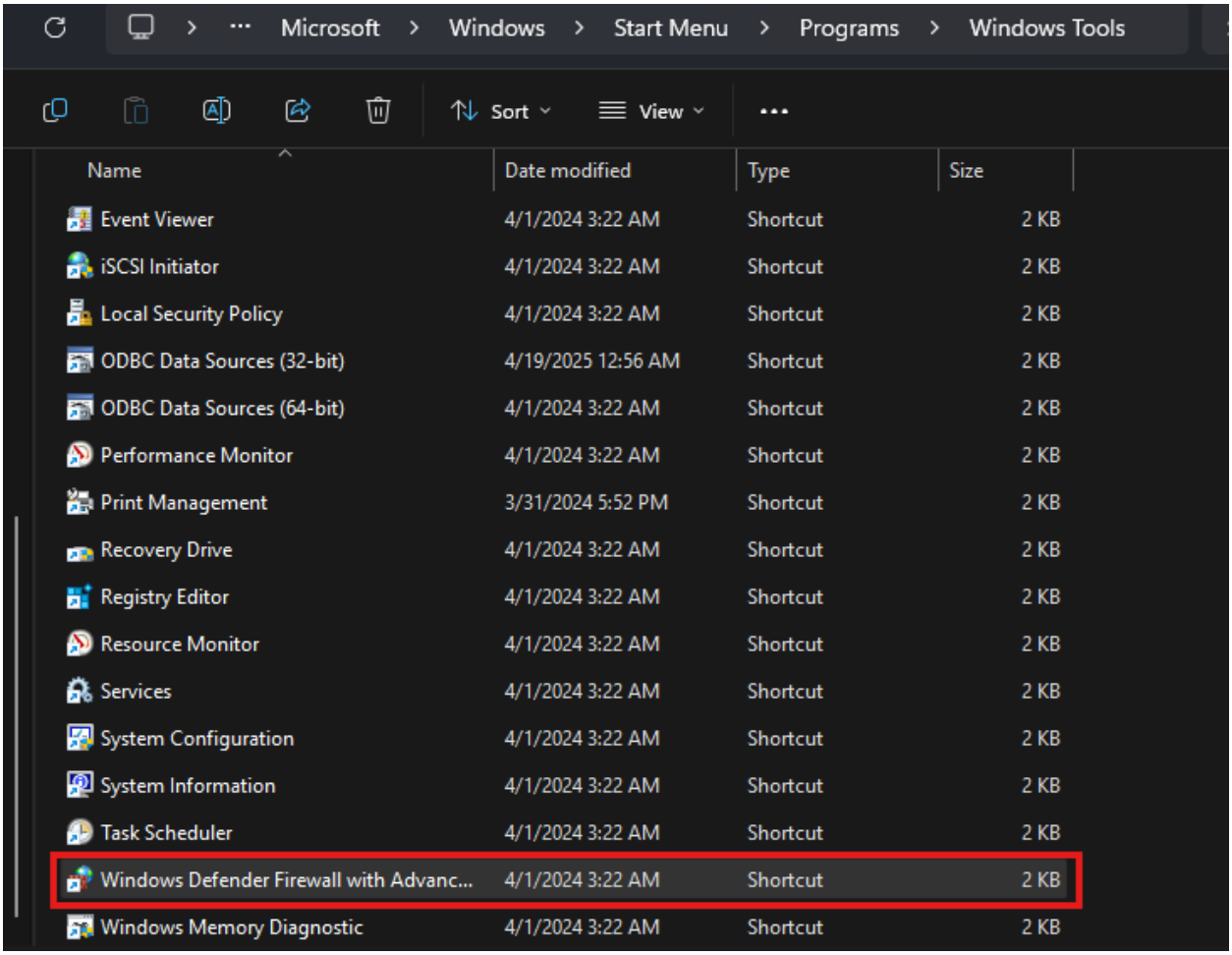
1. In File Explorer, go to file path **C:\inetpub\wwwroot**
2. Create a new folder **CertEnroll**



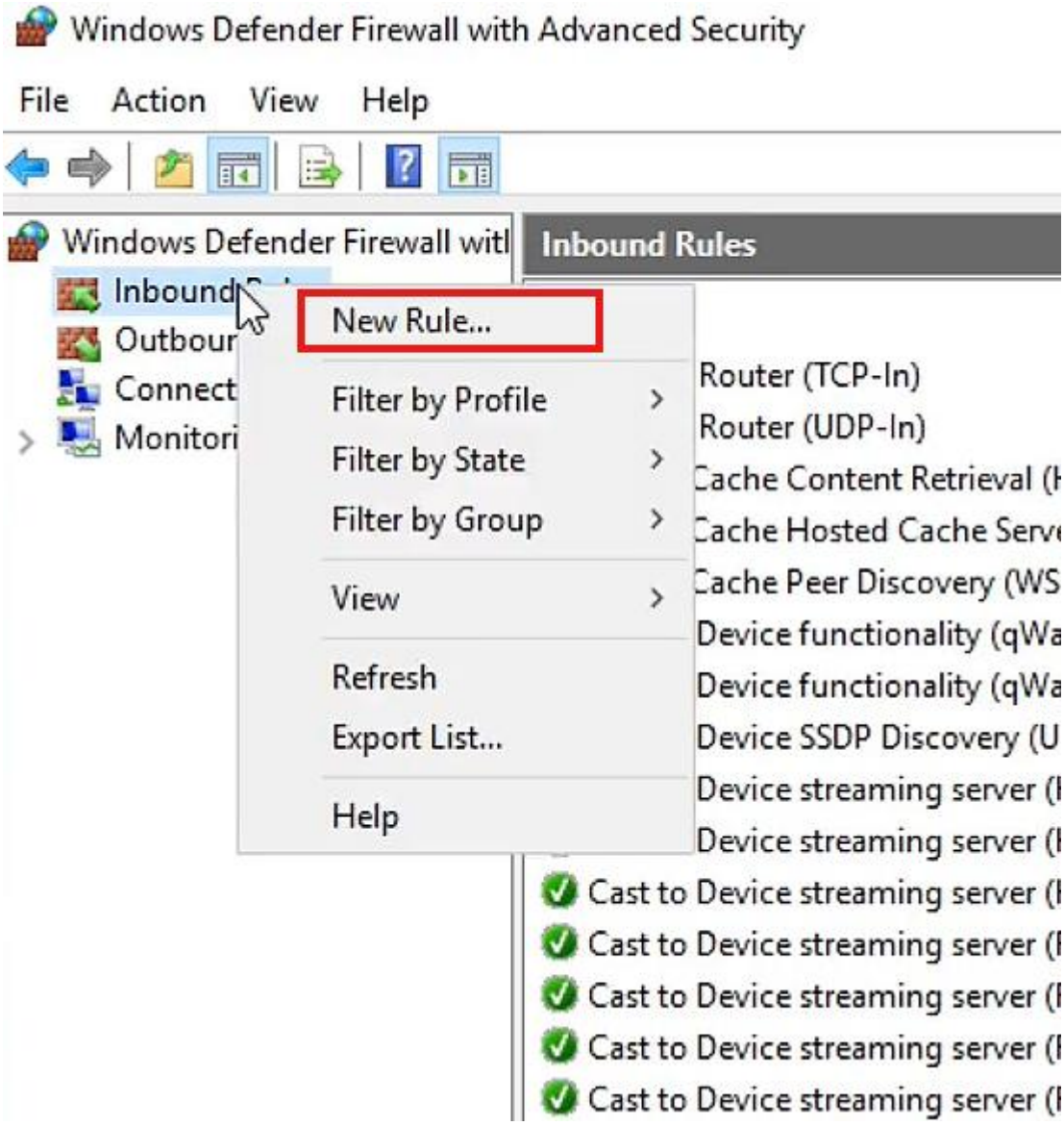
   a. Copy over the root certificate file (.cer) and the root revocation list file (.crl) from C:\Windows\System32\CertSrv\CertEnroll
   b. Ensure the CertEnroll folder has Read access for the IUSR account or Everyone
3. In IIS Manager for our server
   a. **Right-click** the **CertEnroll** folder and select **Convert to Application**
   b. **Ensure .crl** and **.cer** are registered **MIME** types
      i. Right-click server name and open **MIME Types**
      ii. If no **application/x-x509-ca-cert** entry, add it manually
      iii. If no **application/pkix-crl** entry, add it manually
   c. Go to **Request Filtering > Edit Feature** Settings, and check **Allow Double Escaping**

Inbound Firewall Rule to allow PKI Traffic

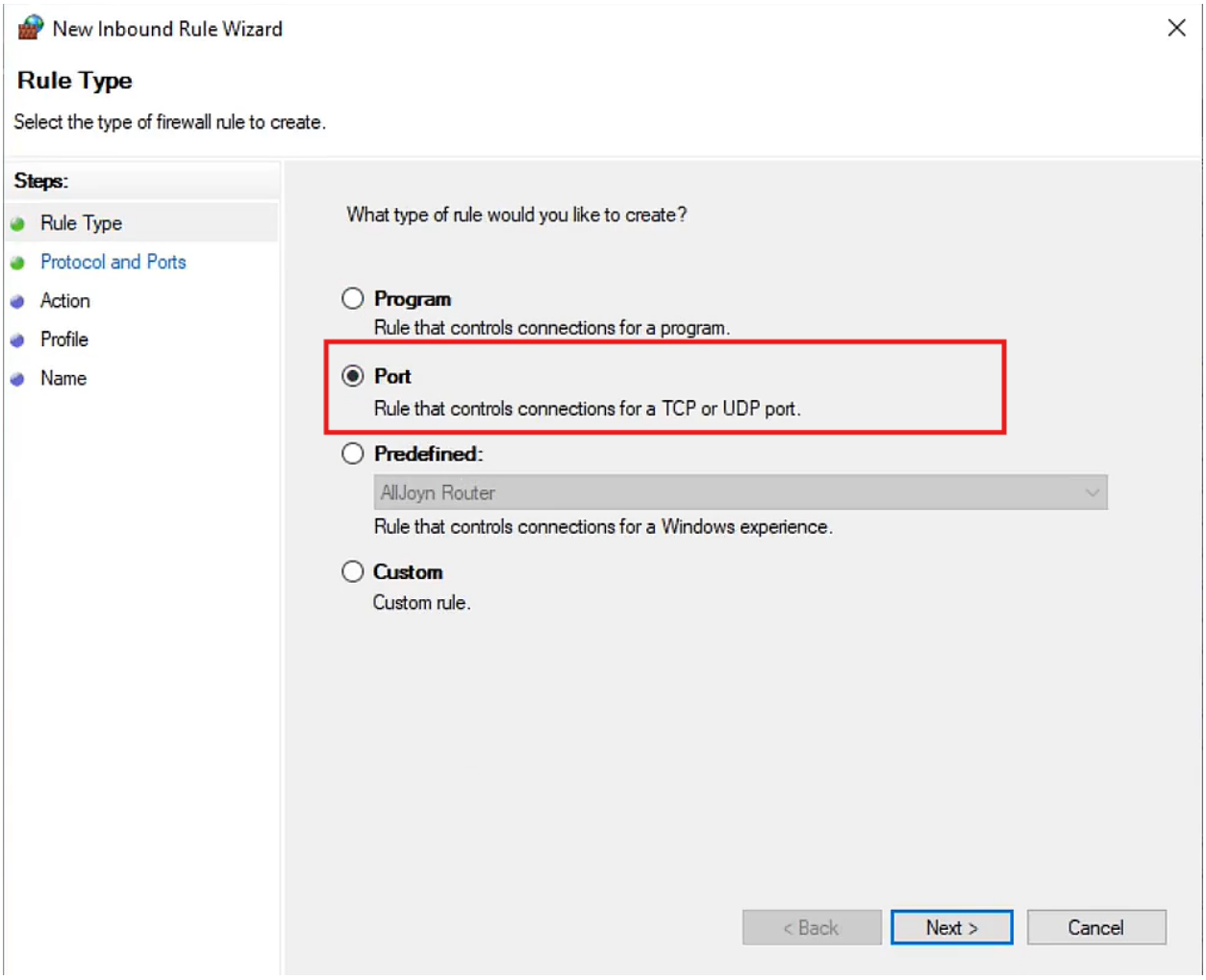
1. Open **Windows Defender Firewall with Advanced Security**



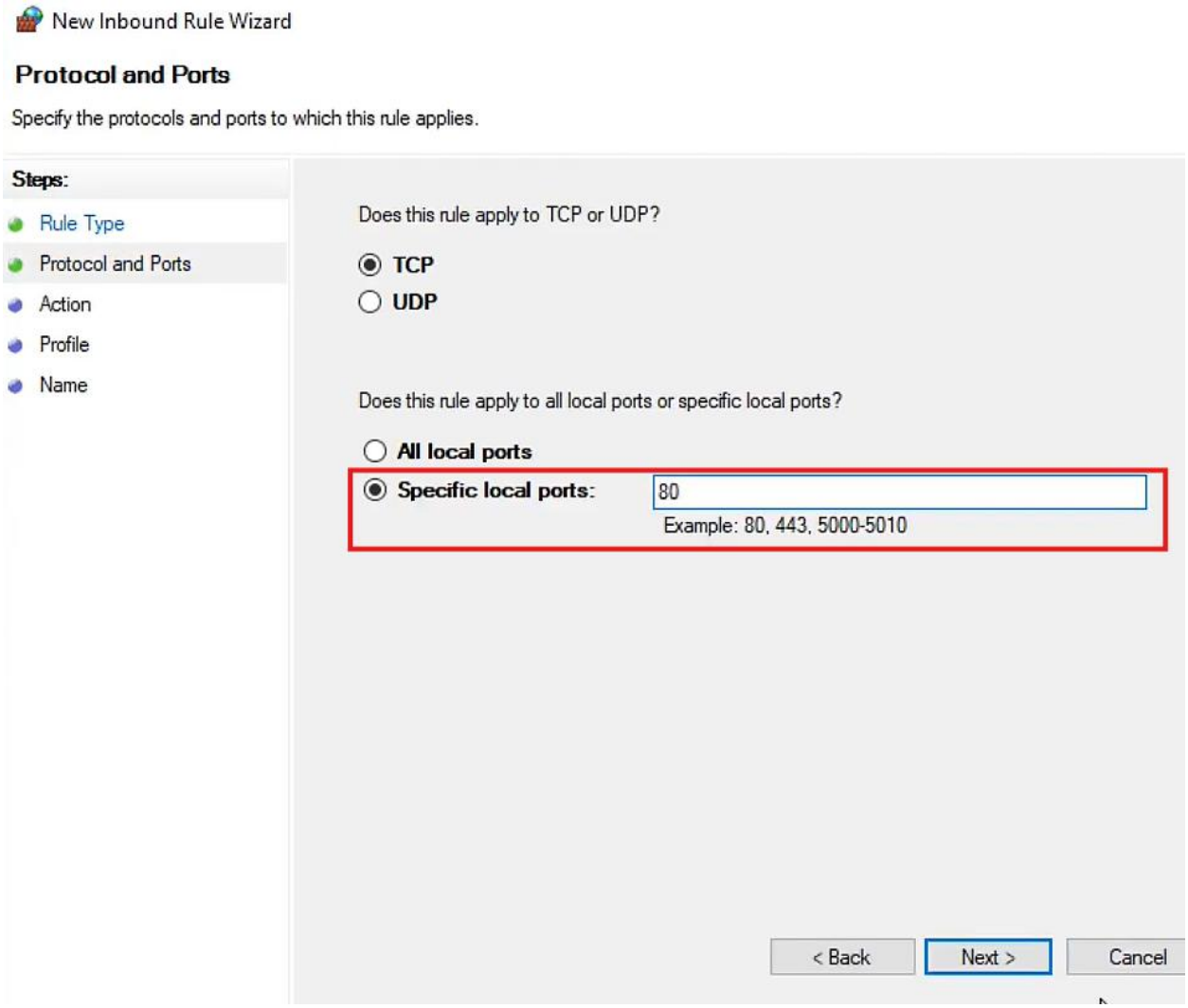
2. **Right-Click Inbound Rule** and select **New Rule...**



3. For **Rule Type**, set as **Port**

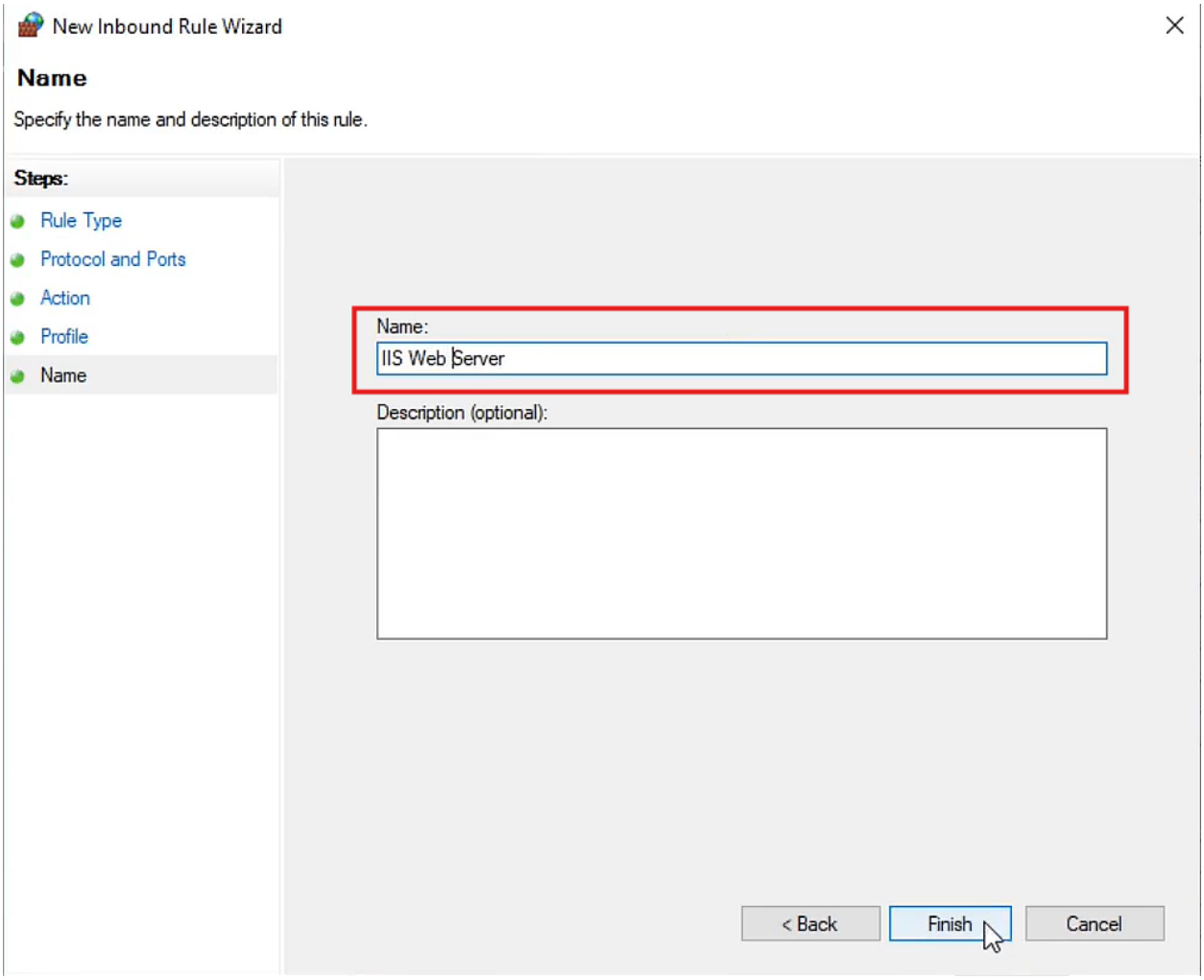


4. Set the **Specific local port** as **80**



5. Accept all defaults and click next

6. Set the **Rule Name** as **IIS Web Server**

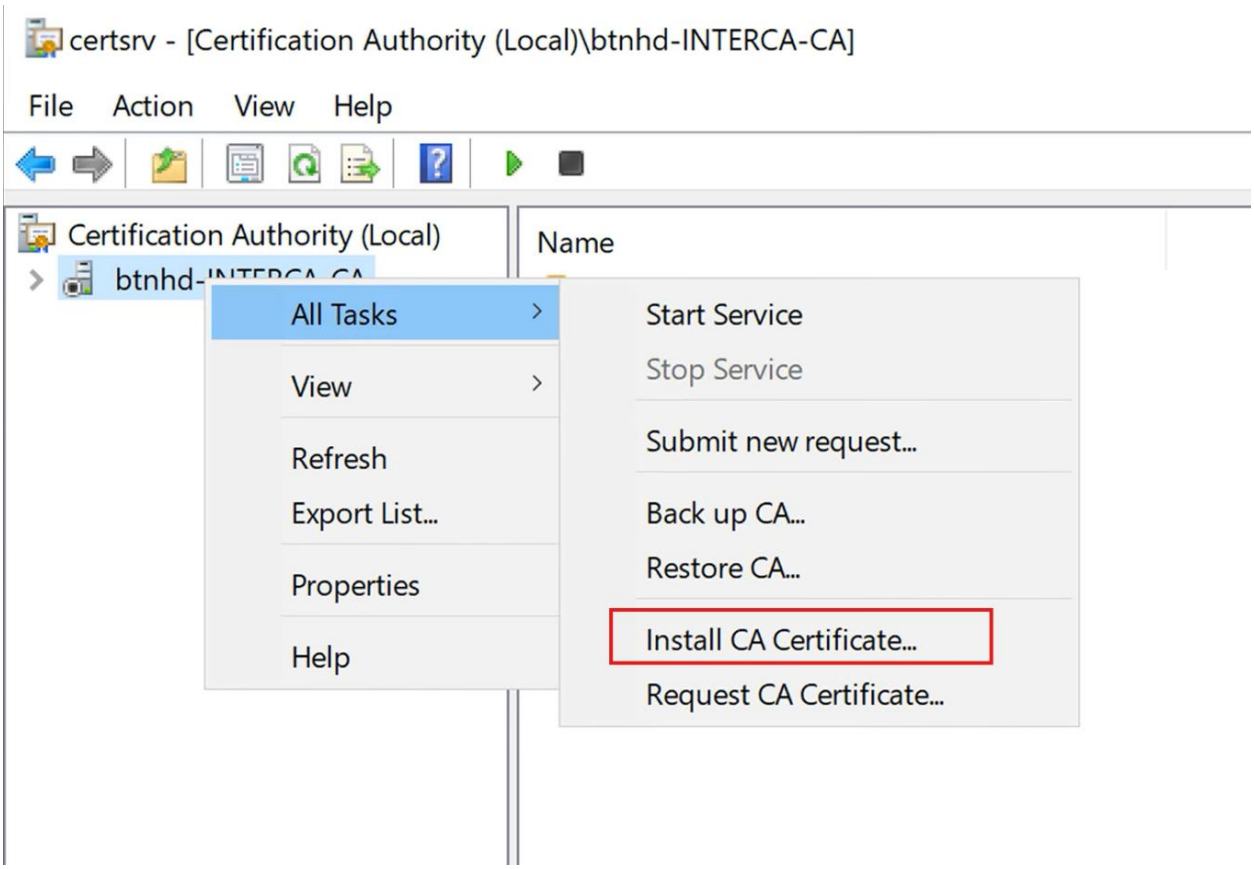


Exporting Intermediate CA Server's Certification Request for Root CA signing

1. Locate the **.req** file (should be in **C:** directory)
2. Transfer the **.req** file into secure media and mount the media onto the Root CA server
3. Refer to SOP for signing the Intermediate CA Server's Certification Request on the Root CA Server

Importing Signed Intermediate CA Server's Certification Request

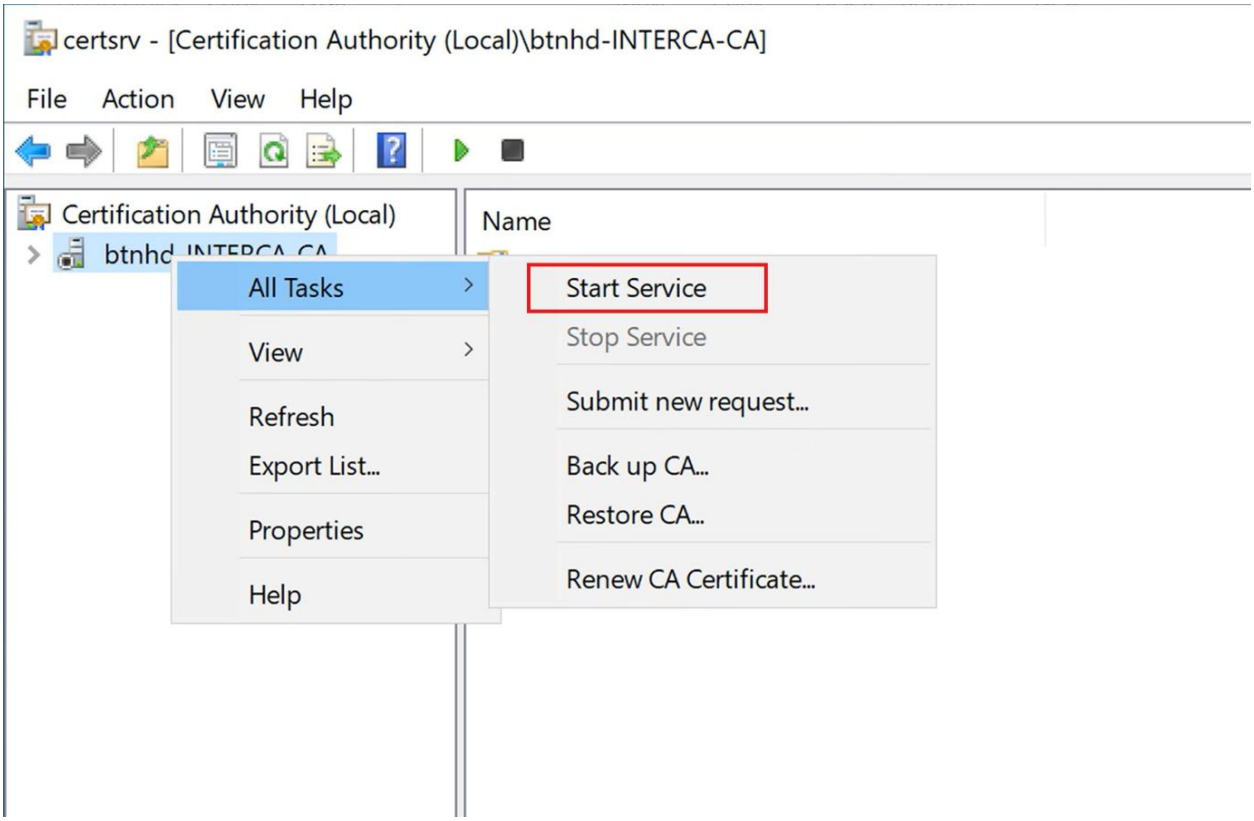
1. Mount the secure media containing the signed **.cer** file
2. Open **Certificate Authority (certsrv.msc)**
3. **Right-click** our server, go to **All Tasks**, and click on **Install CA Certificate...**



4. In the **Select file to complete CA installation** pop-up, go to secure media and **Open** the **InterCA.cer** file
    a. You may need to change the file type filter from *.p7b
    b. **Ensure .cer file is also in the C:\inetpub\wwwroot\CertEnroll folder**
    c. Ensure a copy of the intermediate CRL exists in the same folder as well (may have to publish the CRL to move)

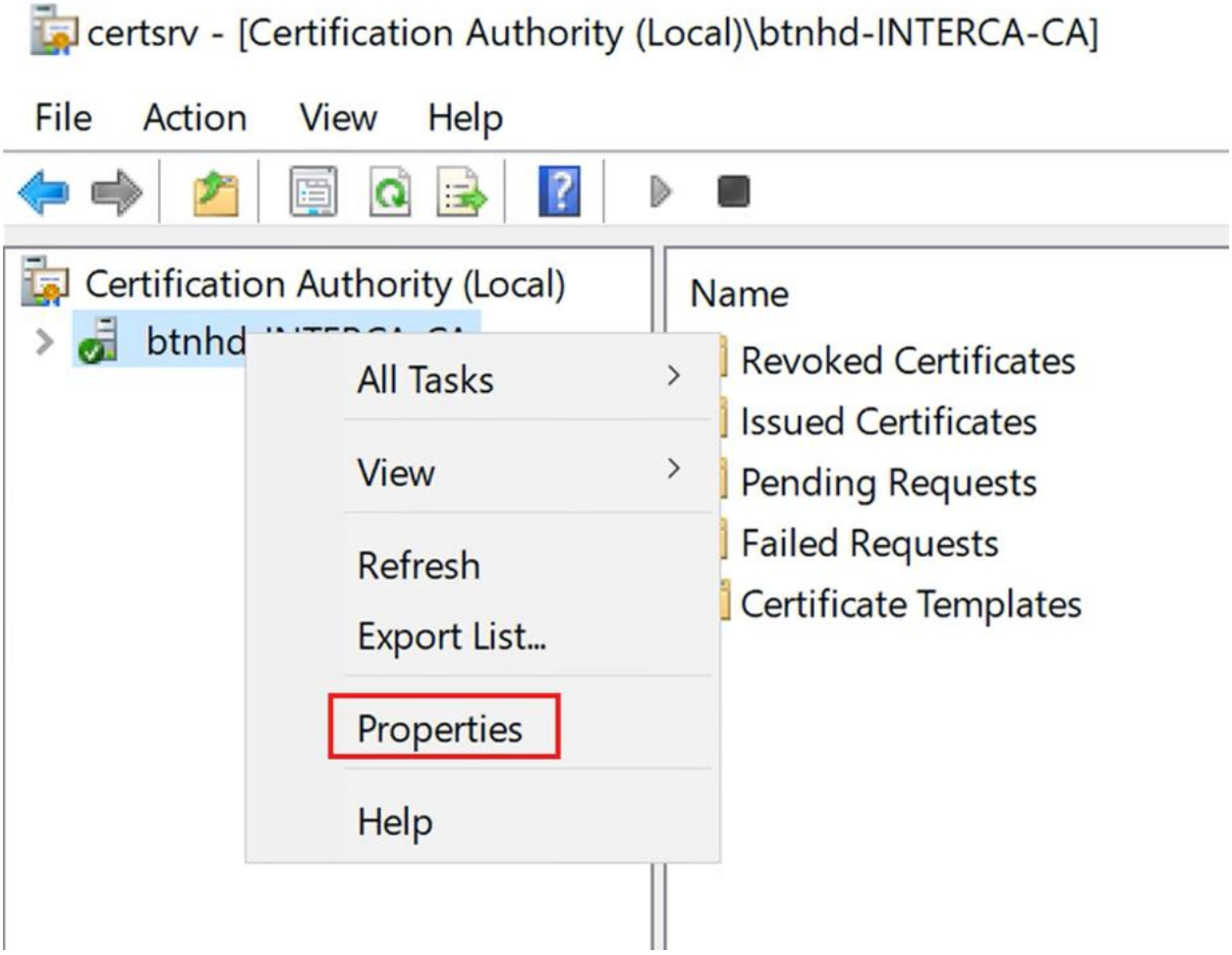
Verification of Successful Root CA Certification Implementation

1. In **Certificate Authority (certsrv.msc)**, **Right-click** our server, go to **All Tasks**, and click **Start Service**

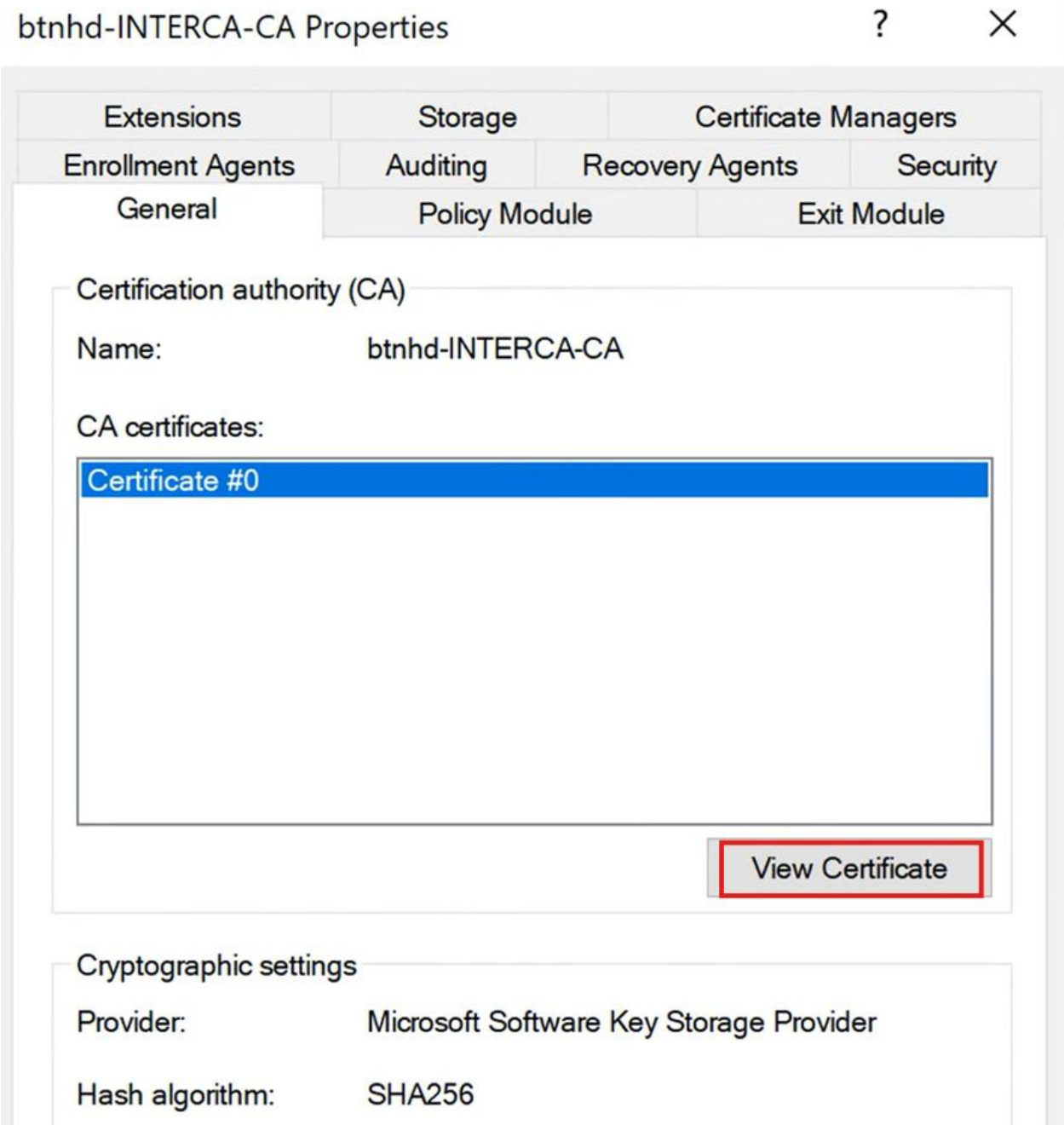


Green checkmark next the server name is already a good sign

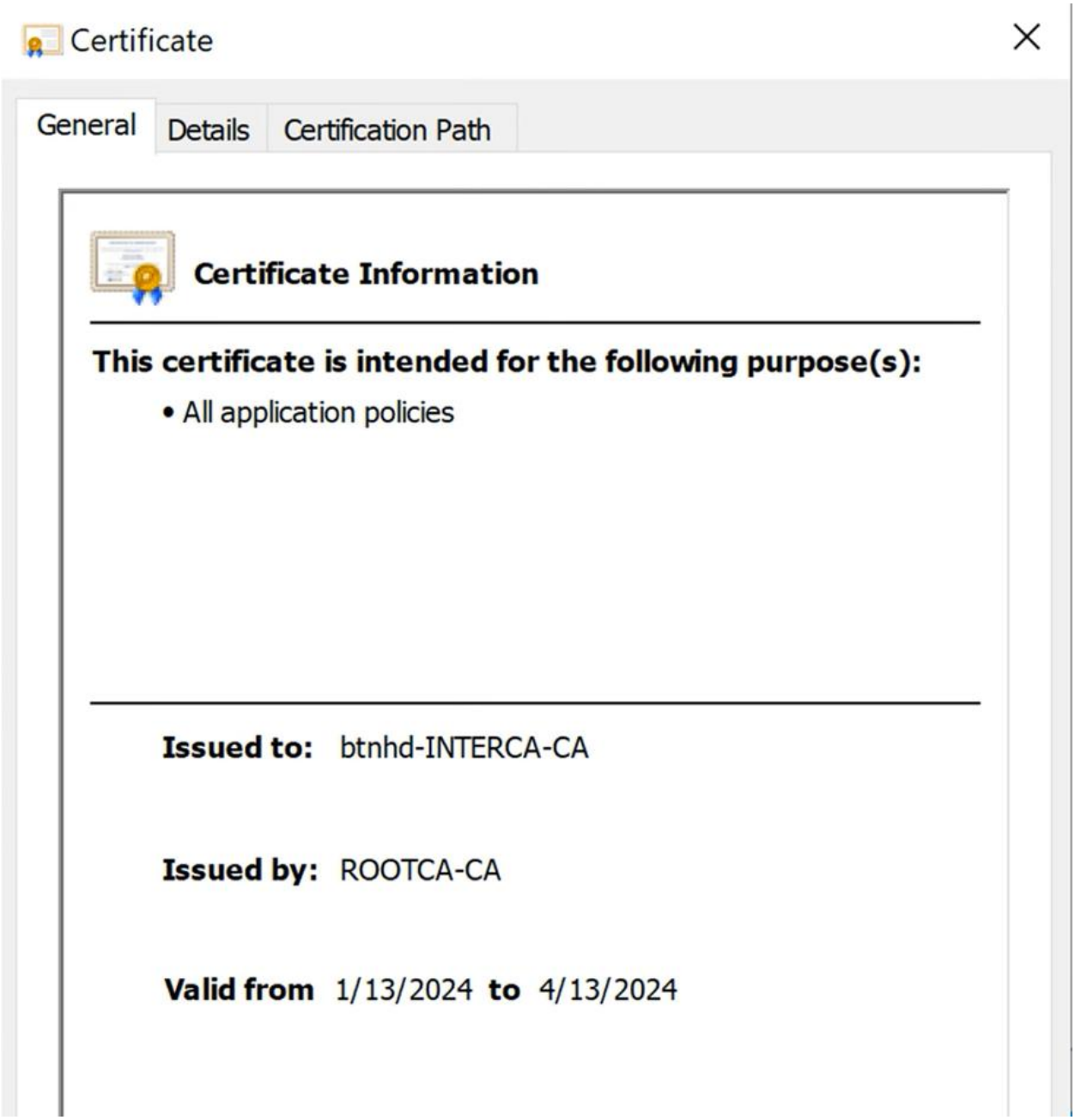
2. For further review of our successful certificate, **right-click** the server and click **Properties**



3. In the **Properties** dialog box, click **View Certificate** for the only available CA Certificate



4. You may review the **Certificate** information to confirm all fields are correct



Clearing any old Certificate Data on INTERCA

If server has any existing certificate data that will be overwritten

1. Run **certlm.msc** (Local Machine Store)
    a. Delete old root from Trusted Root Certification Authorities
    b. Delete old intermediate from Intermediate Certification Authorities
2. Purge the AIA/CDP URL cache
    a. Run as admin: **certutil -urlcache * delete**
3. Clean Up Web Folder
    a. Delete all files in **C:\inetpub\wwwroot\CertEnroll**

Verifying IIS connection

1. Ensure both **.cer** and **.crl** files are in **C:\inetpub\wwwroot\CertEnroll**
2. On a different machine connected to [INTERNAL-ORG], go to URLs:
    a. http://[IIS-Server-Name-Or-FQDN]/CertEnroll/[Cerfile].cer
    b. http://[IIS-Server-Name-Or-FQDN]/CertEnroll/[Crlfile].crl
3. Successful connectivity if both results in opening/downloading file

Verifying IIS validity

1. Ensure both **.cer** and **.crl** files are in **C:\inetpub\wwwroot\CertEnroll**
2. Open command prompt
    a. certutil -url http://<Your-IIS-Server>/CertEnroll/<Your-INTERCA-Name>.crl
3. In tool window, select **URL** radio button and click **Retrieve**
4. Successful validity if row is green and reports **Verified**

Next step is to create a Certificate Template on the Domain Controller. There is a SOP for this.